15 Prove algebraically that $0.7\dot{3}$ can be written as $\frac{11}{15}$

Let x = 0.73333... (the 3 repeats).

Then 10x = 7.33333... and 100x = 73.33333...

Subtract: 90x = 66, so x = 66/90 = 11/15.

2 marks total
Shift matching repeat
Subtract to get 90x=66
11/15

(Total for Question 15 is 2 marks)

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